Giridhar Reddy Karnati

kgiridharreddy2022@gmail.com
giridharreddyk.github.io · linkedin.com/in/kgiridharreddy
Greater Hyderabad Area, India

RF and microwave instrumentation · wide-band antenna design and EM simulation · radio receiver characterisation · field-deployable ground stations for satellite telemetry.

EDUCATION

Sep 2022 - Jul 2026

B.E., Electronics & Communication Engineering; Minor in Computer Science

Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad

- CGPA 9.16 / 10
- Academic Excellence Award (Silver Medal) for the highest CGPA up to Semester II (AY 2022-23) in B.E. (ECE-1)
- Coursework: Antennas & Wave Propagation, Microwave Engineering, RF & Microwave Circuits, Electromagnetic Theory, Communication Systems, Digital Signal Processing, Digital System Design (Verilog)
- Major project: Design of Anti-Spoofing Antenna for Electronic Warfare (guide: Dr. K. Suman)

2020 - 2022

Intermediate (Class XI-XII), Mathematics-Physics-Chemistry

Sri Chaitanya Junior Kalasala, Hyderabad

- Aggregate: 96.2%

2020

Class X

VijayaRatna High School

- CGPA: 10 / 10

RESEARCH & PROFESSIONAL EXPERIENCE

Jul 2026 - present

NPTEL Pre-Doc Fellow (Project Associate)

IC Design Lab, Dept. of Electrical Engineering, IIT Hyderabad · Advisor: Prof. Abhishek Kumar

Front-end for a wideband receiver in CMOS.

May 2026 - Jul 2026

Facility Technologist Intern

National Nanofabrication Centre (NNfC), CeNSE, IISc Bengaluru

Nov 2025 - May 2026

Visiting Research Student

Astronomy & Astrophysics, Raman Research Institute (RRI), Bengaluru

Architectural design and bring-up of a water-vapour radiometer; RF link-budget and thermal-noise analysis, and gain / noise-figure cascade modelling of the IF chain.

Aug 2025 - Sep 2025

Student Intern, Nanofabrication Lab (SPARC NANO-X)

IIT Hyderabad

Clean-room processing: laser lithography, sputtering, reactive-ion etching (RIE) and plasma cleaning. Trained on the IC design flow from layout through packaging; fabricated nano-silicon waveguides, microfluidic devices and 50 µm Au interdigitated electrodes for 2-D-material gas sensors using Heidelberg µPG 501 maskless lithography, DC magnetron sputtering and lift-off in a Class-1000 clean room.

Jul 2025 - Aug 2025

IEEE AP/MTT-S Summer Intern, RF & Microwave Circuits

IIT Kharagpur (remote) · Mentor: Shrawan Kumar Patel

RF and microwave circuit analysis and synthesis: designed and simulated microwave phase shifters and coupled-line couplers in Keysight ADS, with MATLAB models for comparison; evaluated insertion loss, phase error and return loss across the band.

Jul 2025

Electrical Engineering Summer School (EESS-25)

Indian Institute of Science (IISc), Bengaluru

Selected as one of 100 students nationwide for an intensive programme in signal processing, communications and microelectronics.

Jun 2025 - Jul 2025

Project Intern, SDR & Embedded Radio Systems

National Institute of Technology Karnataka (NITK), Surathkal

Built SDR-based transceivers and Raspberry Pi-driven portable APRS ground nodes for weather monitoring, deployable to remote field sites.

Mar 2025 - May 2026

Internship Trainee, Antenna Design & RF Hardware (part-time)

RaoS Consultants LLC (remote)

Comprehensive Internship Course on Antenna Design (CICAD 2025-26): electromagnetics, antenna theory, horns, reflectors and phased arrays, design and analysis, software and hardware, integration and qualification, measurements and testing. Tools: CST Microwave Studio. One of 24 students selected from 90+ applicants on the basis of CGPA and motivation, following the February 2025 announcement; trained through monthly online sessions. Joint 2nd Place Best Intern, based on assignments.

Learned from Dr. Sudhakar Rao, Prof. Sameer Chakravarthy Vedula, Prof. Prabhakar Pathak, Dr. C. J. Reddy, Prof. Satish Sharma, Dr. Nacer Chahat, Dr. Gaurangi Gupta and Ms. Sriya Kotta.

Dec 2024 - Jan 2025

Technical Intern

India Space Week (remote)

Oct 2024

Research Trainee, ICAN 4.0 Workshop

CREST, BITS Pilani Hyderabad Campus

Fabricated and electrically characterised 2-D-material-based nano-device sensors.

Jun 2024 - Jul 2024

Internship Trainee

National Institute of Electronics & Information Technology (NIELIT)

PUBLICATIONS

2025

VayuVani: A Self-Managed LoRa Ground Station for Amateur Operators

K. Giridhar Reddy · HAM Fest India 2025 Magazine, Goa, December 2025, pp. 20-27

2025

Design of Ground Station Antennas for Tracking of LoRa-Based Satellites at 433 MHz Amateur Frequencies

K. Vasanth, **K. Giridhar Reddy**, V. Bala Sai Datta, K. Suman · Proc. IEEE SCEECS 2025, MANIT Bhopal, January 2025 · IEEE Xplore Best Paper Award

2024

Privacy-Preserving LoRa Ground Stations in Amateur Satellite Communications: A Case Study of VayuVani

K. Giridhar Reddy · Radio Amateurs' World (RAW) Magazine · HAM Fest India 2024, Kolkata, 21 December 2024

VayuVani is an open-source LoRa ground-station platform created by Chandra Tungathurthi (vayuvani.com). My articles and talks describe building and operating ground stations with it.

HONOURS & AWARDS

Aug 2026

IEEE SPACE Young Professional (YP) Laureate

IEEE SPACE 2026 (Space, Aerospace and Defence Conference)

Jun 2026

Best Intern Award, 2nd Place

CICAD 2025-26, RaoS Consultants LLC · joint second place

The announced second-place award package included ₹6,000, a plaque, a certificate, a travel grant and accommodation, with presentation scheduled for the WAMS 2026 Banquet Awards Ceremony.

2025 & 2026	**GATE qualified, Electronics & Communication Engineering** GATE 2025 (score 436) and GATE 2026
Jan 2026	**Dayton Amateur Radio Association Scholarship** ARRL Foundation, USA · featured in QST, ARRL's monthly journal
Dec 2025	**B.Tech Initiative Travel Grant** IEEE Microwaves, Antennas and Propagation Conference (MAPCON 2025), Kochi
Dec 2025	**Rank 20 of 250 (top 8%), AI/ML Powered FPGA Design & Simulation Hackathon** National Institute of Technology Jamshedpur
Jul 2025	**Selected for the IISc Electrical Engineering Summer School (EESS-25)** Top 100 nationwide
Mar 2025	**1st Place, Paper Presentation, VANSH-2K25** Vignan Institute of Technology and Science, Deshmukhi, Hyderabad
Mar 2025	**1st Place, Poster Presentation (Pradarshana), Aakruthi 2025** National Technical Symposium, Dept. of ECE, University College of Engineering, Osmania University, Hyderabad
Jan 2025	**Best Paper Award (ECE)** IEEE SCEECS 2025, MANIT (NIT) Bhopal, 18-19 January 2025
Jan 2025	**2nd Place, Quiz Competition, NAFED08** VSSC / ISRO, held at IIT Hyderabad
Mar 2024	**Academic Excellence Award (Silver Medal)** CBIT, Institute Day (SHRUTHI-2024) · highest CGPA up to Semester II (AY 2022-23) in B.E. (ECE-1)
2024	**7th Place, RBI90Quiz 2024 State-Level Round** Reserve Bank of India (RBI) Represented CBIT with teammate Bala Sai Datta Veturi; placed seventh in the state-level round and narrowly missed advancing to the next round.
2024	**Three-time NPTEL Star** Discipline Star (Computer Science & Engineering) and Believer Star, Jan-Apr 2024 · Enthusiast Star, Jul-Dec 2024

TALKS

Oct 2025	**VayuVani: your private LoRa satellite ground station** HamFest India 2025, Goa College of Engineering, Farmagudi, Goa
Dec 2024	**Speaker, 7th Lamakaan Annual Radio Convention** Lamakaan Amateur Radio Club, with the Dept. of ECE & IEEE, Muffakham Jah College of Engineering & Technology, and the Digital Empowerment Foundation, Hyderabad

LEADERSHIP & OUTREACH

Oct 2023 - Apr 2026

Founder & Student President, then Senior Student Advisor

HAM Radio Club of CBIT, Hyderabad

- Founder & Student President, Oct 2023 - Nov 2025; Senior Student Advisor, Oct 2025 - Apr 2026.
- Led technical projects and STEM-outreach workshops for school students.
- Two-day workshop with the Dept. of ECE, “Design approaches of Antennas for HAM Radio Applications using Open Source Software” (26-27 April 2024): antenna basics, modelling and simulation in 4NEC2, planar antenna design and realization, Morse code, cables and connectors, amateur-radio regulations and licensing, and practical transmitter, receiver and radio communication applications.
- Conducted hands-on HAM radio sessions for Class 10 students at ZPHS Nadergul, Ranga Reddy district, as part of a COPE (Community of Practice for Education) initiative. Introduced amateur radio, its applications and advantages, emergency communications, and its role in science, technology and STEM learning.
- School-outreach acknowledgements: Akhil Reddy, Geethanjali Kamoji, Jhansi Nomula, Sanjana, Dr. Muddamalla Naresh and A. S. Keerthi Nayani for organizing and supporting the initiative.

2025

Technical Chair & Coordinator, SYNAPSE 2025

CBIT · managed all technical events for the college techno-cultural festival

Dec 2024

Volunteer and contributor, STEM and COPE workshops for school children

IEEE Microwaves, Antennas and Propagation Conference (MAPCON 2024), Hyderabad

PROFESSIONAL MEMBERSHIPS

Jun 2025 - present

Member

Applied Computational Electromagnetics Society (ACES)

May 2025 - present

Individual Member

RISC-V International

Jan 2025 - present

Associate Member

NoRCEL Institute (Network of Researchers on the Chemical Emergence of Life)

Nov 2024 - present

Student Member

Institute of Electrical and Electronics Engineers (IEEE)

Sep 2024 - present

Student Member

Radio Amateur Satellite Corporation (AMSAT)

Active student member supporting amateur radio in space and satellite communications.

Aug 2023 - Apr 2026

ISF Member (IETE Students' Forum)

Institution of Electronics and Telecommunication Engineers (IETE)

LICENCES & CERTIFICATIONS

Feb 2024

Amateur Wireless Station Licence, General Grade (callsign VU2TIO)

Department of Telecommunications, Government of India

Credential ID / callsign: VU2TIO

Sep 2026

IEEE MTT-S Virtual Workshop on AI for Microwaves

IEEE Microwave Theory and Technology Society

Nov 2025, May 2026

IEEE Certificates Program (CEU/PDH), two certificates

IEEE Educational Activities · topics include generative AI and 5G

November 2025 credential ID: 166247989

Mar 2026	**International Physics Competition (IPhyC) 2025, semi-final round** Participation certificate
Dec 2025	**International Youth Math Challenge (IYMC) 2025, qualification round** International Youth Math Challenge · participation certificate Credential ID: QR-2025-100BAC997467
Feb 2024	**Certificate of Merit, Geo-data Sharing and Cyber Security** Indian Institute of Remote Sensing (IIRS), Indian Space Research Organisation (ISRO) Credential ID: 4f0e476de55d5c526e75b62a9b1ad866
Aug 2025	**Summer Internship Programme 2025** Indian Institute of Technology Kharagpur Credential ID: cdec623d-f077-4268-abbc-6d29aa8eeb56
Dec 2024	**NPTEL Enthusiast Star · Jul-Dec 2024** NPTEL Credential ID: NPTELSS2240103728
May 2024	**NPTEL Believers Star · Jan-Apr 2024** NPTEL Recognized for completing four courses in one semester. Credential ID: NPTELSS1240100661
May 2024	**NPTEL Discipline Star in Computer Science & Engineering · Jan-Apr 2024** NPTEL Recognized for exceeding 56 weeks of cumulative learning in the Computer Science and Engineering discipline. Credential ID: NPTELSS1240103455
Jun 2023	**Graphic design essentials** Canva Credential ID: d572e7

TECHNICAL SKILLS

EM & RF simulation	Ansys HFSS, CST Studio Suite, Altair FEKO, Keysight ADS, MATLAB RF Toolbox
RF & microwave	Wide-band antennas, LNA, mixer, PLL/VCO, power amplifier, filter design, link-budget analysis, noise figure / IIP3, anechoic-chamber and VNA measurement workflows
RF hardware & bench	VNA, spectrum analyser, SDR (RTL-SDR, HackRF), soldering and RF connectorisation, PCB design and prototyping
Digital, VLSI & EDA	Cadence Virtuoso, Xilinx Vivado, Silvaco TCAD (Victory Process / Device / Extract), CleWin, SolidWorks
Programming & HDL	Python, C/C++, MATLAB, Verilog, VHDL, LabVIEW
Embedded	Arduino, Raspberry Pi, ESP32 (Xtensa LX6), RISC-V, Arm Cortex-M, FPGA prototyping
Fabrication	Photolithography, maskless laser writing, DC magnetron sputtering, reactive-ion etching (RIE), plasma cleaning, PVD, CVD, lift-off, RTA; I-V characterisation (Keithley 2450)
Languages	English (professional working proficiency), Hindi (native / bilingual), Telugu (native / bilingual)